Forcing Reading Group — Exercise Sheet 2

The theory $I\exists_1(f)$

Definition 1. We define $\exists_1(f)$ to be the set of all $L_{PA}(f)$-formulas of the form $(\exists y_1)(\exists y_2)\dots(\exists y_k)\varphi(x_1,\dots,x_k,y_1,\dots,y_k)$, where φ is quantifier-free and f is a newly introduced function symbol. The scheme $IND(\exists_1(f))$ consists of

$$(\varphi(0)\wedge(\forall x)(\varphi(x)\to\varphi(x+1)))\to(\forall x)(\varphi(x))$$

for every $\varphi\in\exists_1(f)$. The scheme $LNP(\exists_1(f))$ consists of

$$(\exists x)(\varphi(x))\to(\exists x)(\varphi(x)\wedge(\forall y<x)(\neg\varphi(x)))$$

for every $\varphi\in\exists_1(f)$.

Exercise 2. Show that $\mathrm{Th}(\mathbb{N})+LNP(\exists_1(f))\vdash IND(\exists_1(f))$.

Definition 3. $I\exists_1(f)=B+IND(\exists_1(f))$ for a sufficiently strong $B\subseteq\mathrm{Th}(\mathbb{N})$.

Exercise 4 (*)**.** Does $\mathrm{Th}(\mathbb{N})+IND(\exists_1(f))\vdash LNP(\exists_1(f))$?

Exercise 5 (*)**.** Does Q suffice as B to prove the previous exercises with $\mathrm{Th}(\mathbb{N})$ replaced by $I\exists_1(f)$?

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Remark 6. In this section we will fix a countable nonstandard $M\models\mathrm{Th}(\mathbb{N})$ and $n\in M\setminus\mathbb{N}$.

Exercise 7. Let $PHP(f)$ be the formula

$$(\forall x)((\exists y\le x)(f(y)\ge x)\vee(\exists y_1\le x)(\exists y_2\le x)(f(y_1)=f(y_2))).$$

Try to explicitly define a function $f:\{0,\dots,n+1\}\to\{0,\dots,n\}$ such that $(M,f)\models\neg PHP(f)$. Does your $(M,f)\models I\exists_1(f)$?

Definition 8. Let P be the set of all (standardly) finite partial injections from $\{0,\dots,n+1\}$ to $\{0,\dots,n\}$. For all $p,q\in P$ we define $q\le p\Leftrightarrow q\supseteq p$.

Exercise 9. Every (standardly) finite set of pairs of elements of M can be represented by a single element of M by a suitable encoding. Show that P, interpreted as a set of codes, is not definable in M.

Remark 10. Our goal is to build a more sophisticated f such that $(M,f)\models I\exists_1(f)+\neg PHP(f)$. We will build f as a union of a chain $p_0\ge p_1\ge p_2\ge\dots$ in P. We will pick the p_i's such that:

- Every pigeon $y\in\{0,\dots,n+1\}$ is mapped somewhere.
- Every $\exists_1(f)$-formula φ satisfies $LNP(\varphi)$.

Exercise 11. Define a relation $\Vdash$ between P and quantifier-free $L_{PA}(f)$-formulas such that: $p \Vdash \varphi \Leftrightarrow \forall f \supseteq p : (M, f) \models \varphi$.

Exercise 12. Extend the previous relation to $\exists_1(f)$-formulas.

Exercise 13. Show that the previous relation cannot be extended to $\forall_1(f)$-formulas.

Exercise 14. Show that for every $\exists_1(f)$-formula φ and every $p \in P$, the set

$$\{a; \exists q \leq p : q \Vdash \varphi(a)\}$$

is definable in M.

Exercise 15. Construct an f such that $(M, f) \models I\exists_1(f) + \neg PHP(f)$.